



LESSON - 15



SIKHOGE KYA???

INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

SQL STATEMENT FUNDAMENTALS

- SELECT * FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- UNION, UNION ALL, MINUS, INTERSECT
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES

Aggregate Functions

SQL Aggregate Functions are essential for summarizing and analyzing large datasets. These functions operate on multiple rows of a table, combining them into a single, more meaningful result. SQL Aggregate functions are mostly used with the GROUP BY clause of the SELECT statement.

Common Aggregate Functions

SUM() – Returns the total sum of a numeric column.

AVG() – Returns the average value of a numeric column.

COUNT() – Returns the number of rows that match the query.

MAX() – Returns the maximum value in a column.

MIN() – Returns the minimum value in a column.

GROUP_CONCAT() (MySQL specific) – Concatenates values from a group into a single string.

Aggregate Functions Example

--Count the number of employees

```
SELECT COUNT(*) AS TotalEmployees FROM Employee;
```

-- Calculate the total salary

```
SELECT SUM(Salary) AS TotalSalary FROM Employee;
```

-- Find the average salary

```
SELECT AVG(Salary) AS AverageSalary FROM Employee;
```

-- Get the highest salary

```
SELECT MAX(Salary) AS HighestSalary FROM Employee;
```

-- Determine the lowest salary

```
SELECT MIN(Salary) AS LowestSalary FROM Employee;
```